THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — MATHEMATICS (Standard) · MID-TERM PRACTICE PAPER

Session 2026-27 · CBSE / Directorate of Education pattern · Sub code 041

Class: 10	Subject: Mathematics	Set 01	Time: 3 Hours	Max. Marks: 80

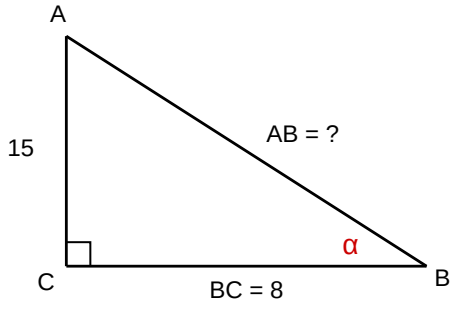
General Instructions:

1. This Question Paper has 5 Sections 'A', 'B', 'C', 'D' and 'E'.
2. Section A has 20 MCQs carrying 1 mark each (Q19–20 are Assertion–Reason questions).
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based questions (04 marks each) with sub-parts of the values of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice in 2 questions of 5 marks, 2 questions of 3 marks and 2 questions of 2 marks has been provided. An internal choice has been provided in the 2-mark questions of Section E. You have to attempt only one option out of the given options.
8. Draw neat figures wherever required. Take π = 22/7 wherever required, if not stated.
9. There is no provision of negative marking. Use of calculator is not permitted.
10. Please write down the serial number of the question before attempting it.

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. The value of sin α from the given figure is **1**



(a) $\frac{8}{17}$ (b) $\frac{15}{8}$ (c) $\frac{8}{15}$ (d) $\frac{15}{17}$

2. D and E are points on the sides AB and AC of ΔABC such that ∠ADE = ∠ACB. Then **1**

(a) ΔADE ~ ΔABC (b) ΔADE ~ ΔACB (c) ΔAED ~ ΔACB (d) ΔAED ~ ΔABC

3. The HCF and LCM of 12, 18 and 30 are respectively **1**

(a) 2, 360 (b) 6, 360 (c) 6, 180 (d) 3, 180

4. Arun and Bela were both born in September 2010. The probability that they have the same date of birth is **1**

(a) $\frac{1}{30}$ (b) $\frac{1}{31}$ (c) $\frac{1}{365}$ (d) $\frac{1}{60}$

5. The distance of a point A from the *y*-axis is 4 units. Which of the following cannot be the coordinates of A? **1**

(a) (−4, 2) (b) (4, −5) (c) (2, 4) (d) (4, 0)

6. The value of $(3 + \sqrt{5})(3 - \sqrt{5}) - (\sqrt{2})^2$ is **1**

(a) 4 (b) 2 (c) 6 (d) $7 - 2\sqrt{2}$

7. $(\sqrt{7})^2 - (2 - \sqrt{3})^2$ equals **1**

(a) $4\sqrt{3}$ (b) $4 - 4\sqrt{3}$ (c) $2\sqrt{3} + 1$ (d) $4\sqrt{3} - 3$

8. Meera saves ₹ 200 in January and every following month she saves ₹ 25 more than the previous month. Her saving in July is **1**

(a) ₹ 325 (b) ₹ 375 (c) ₹ 400 (d) ₹ 350

9. If the system of equations $(a - 1)x + 3y = 6$ and $4x + 6y = 12$ has infinitely many solutions, then a equals **1**

(a) 3 (b) 2 (c) 4 (d) 5

10. Which of the following is true for all values of θ for which the expressions are defined? **1**

(a) $\sin^2\theta = 1 + \cos^2\theta$ (b) $\text{cosec}^2\theta = 1 + \cot^2\theta$

(c) $\sec^2\theta - \tan^2\theta = 0$ (d) $\tan\theta = \cot\theta$

11. The point on the y-axis which is equidistant from A(3, 5) and B(−3, 1) is **1**

(a) (0, 2) (b) (0, 4) (c) (3, 0) (d) (0, 3)

12. A 2-digit number less than 50 is chosen at random. The probability that it is a multiple of 3 and 4 but not of 5 is **1**

(a) $\frac{1}{5}$ (b) $\frac{1}{8}$ (c) $\frac{1}{10}$ (d) $\frac{3}{40}$

13. The solution of the pair of linear equations $3x + 2y = 16$ and $x - 2y = 0$ is **1**

(a) x = 4, y = 2 (b) x = 2, y = 4 (c) x = 6, y = 3 (d) x = 3, y = 4

14. If a and b are two different prime numbers, the HCF of a^3b^2 and a^2b^5 is **1**

(a) a^3b^5 (b) a^2b^5 (c) ab (d) a^2b^2

15. Two dice are thrown together. The probability that the sum of the numbers is a prime number is **1**

(a) $\frac{1}{3}$ (b) $\frac{5}{12}$ (c) $\frac{7}{12}$ (d) $\frac{11}{36}$

16. If $\tan\theta = \frac{8}{15}$, then $\cos\theta$ equals **1**

(a) $\frac{8}{17}$ (b) $\frac{17}{15}$ (c) $\frac{15}{17}$ (d) $\frac{15}{8}$

17. The pair of linear equations $2x - 5y = 3$ and $4x - 10y = 7$ represents two lines which are **1**

(a) coincident (b) intersecting at exactly one point

(c) parallel (d) intersecting at exactly two points

18. The common difference of the AP √12, √27, √48, … is **1**

(a) 2√3 (b) √3 (c) 3√3 (d) 5√3

19. **Assertion (A):** The points (1, 1), (−1, −1) and (−√3, √3) are the vertices of an equilateral triangle. **1**
Reason (R): If three points P, Q and R are collinear, then PQ + QR = PR.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The value of sec θ can be less than 1 for an acute angle θ. **1**
Reason (R): $\sec\theta = 1/\cos\theta$ and $0 < \cos\theta \le 1$ for an acute angle θ.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

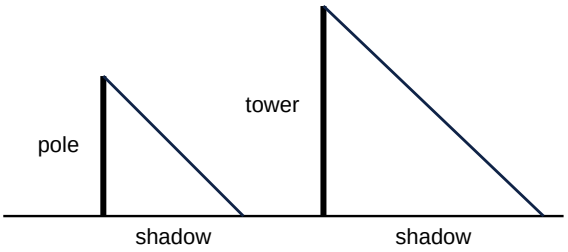
SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Explain why 11 × 13 × 17 + 17 and 5 × 4 × 3 × 2 × 1 + 5 are composite numbers. **2**

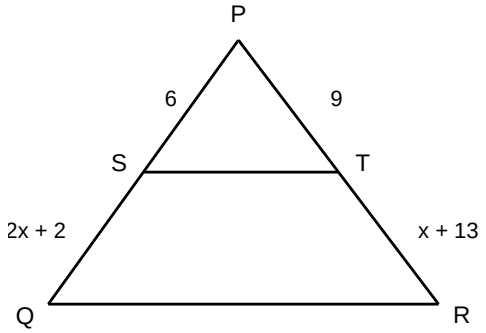
OR

Three bells toll at intervals of 8, 12 and 20 minutes. If they start tolling together, after what time will they next toll together?

22. A vertical pole 5 m long casts a shadow 3 m long on the ground. At the same time a tower casts a shadow 21 m long. Find the height of the tower. **2**

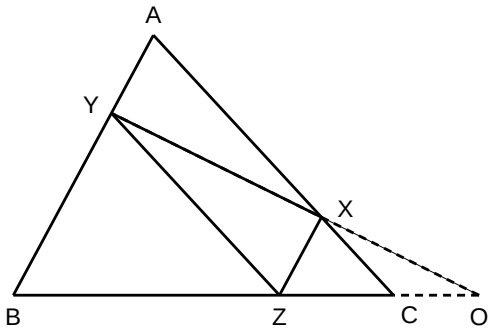


23. In ΔPQR, S and T are points on PQ and PR such that ST ∥ QR. If PS = 6 cm, SQ = (2*x* + 2) cm, PT = 9 cm and TR = (*x* + 13) cm, find *x*. **2**



O R

In the figure, Z is a point on side BC of ΔABC such that XZ ∥ AB and YZ ∥ AC. If XY and CB produced meet at O, prove that $ZO^2 = OB \times OC$.



24. From a point on the ground 24 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is 30°. Find the height of the tower. (Take √3 = 1.732) **2**

25. The HCF of two numbers is 9 and their LCM is 270. If one of the numbers is 54, find the other. **2**

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that 3 − 2√5 is an irrational number, given that √5 is irrational. **3**

27. Solve the pair of linear equations: 99*x* + 101*y* = 499; 101*x* + 99*y* = 501. **3**

28. How many terms of the AP 24, 21, 18, … must be taken so that the sum is 78? Explain the reason for the double answer. **3**

O R

The 8th term of an AP is 31 and its 15th term is 16 more than its 11th term. Find the AP and the sum of its first 11 terms.

29. Find the ratio in which the x-axis divides the line segment joining (3, −6) and (5, 3). Also find the point of division. **3**

OR

Show that A(3, 0), B(4, 5), C(−1, 4) and D(−2, −1) are the vertices of a rhombus.

30. If $\sec\theta = \frac{13}{12}$, evaluate $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$. **3**

31. Cards numbered 21 to 70 are kept in a box and one card is drawn at random. Find the probability that the number is (a) a multiple of 5 (b) a perfect square (c) a prime number less than 33. **3**

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

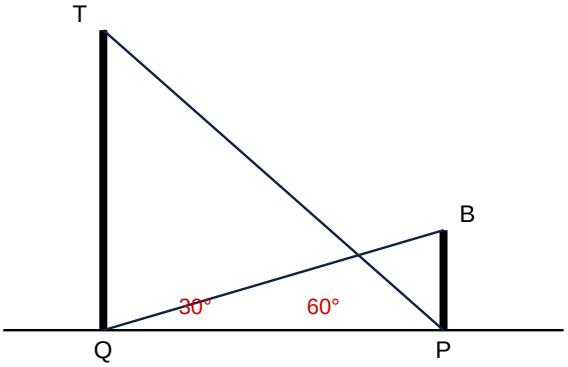
32. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. **5**

33. A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction. **5**

OR

Solve graphically: $2x + y = 8$ and $x - y = 1$. From the graph also find the points where the lines meet the y-axis.

34. The angle of elevation of the top of a building from the foot of a tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the tower is 54 m high, find the height of the building. **5**



35. Prove that $\frac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} = \frac{1}{\sec\theta - \tan\theta}$. **5**

OR

If $\sin(A + B) = 1$ and $\cos(A - B) = \sqrt{3}/2$, $0° < A + B \le 90°$, $A > B$, find A and B. Hence find (i) $\tan A + \cot B$ (ii) $\sec A - \operatorname{cosec} B$.

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. The Class X students of a school organised a tombola at the annual fair. A box has 40 coupons numbered 1 to 40. A participant draws one coupon at random. A multiple of 7 wins a book, a perfect square wins a pen set, and every other draw wins a toffee.

(i) Find the probability of winning a book. 1

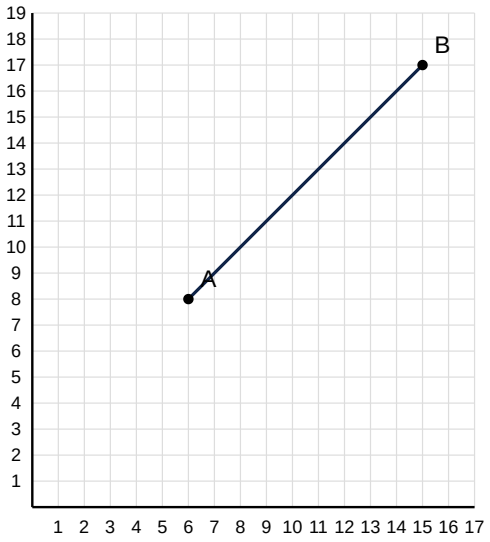
(ii) Find the probability that the coupon drawn is a multiple of 10. 1

(iii) Find the probability that the participant wins a book or a pen set. 2

OR

(iii) The first participant draws coupon 16 and keeps it. Find the probability that the next participant draws a perfect square.

37. The layout of a new housing society is drawn on a coordinate plane with the main gate at the origin O(0, 0). A fountain is at A(6, 8) and the community hall is at B(15, 17), joined by a straight pathway. A bench P is to be placed on AB.



(i) Find the distance of the fountain A from the gate. 1

(ii) Find the coordinates of the midpoint of the pathway AB. 1

(iii) If the bench P divides AB internally in the ratio 1 : 2, find the coordinates of P. 2

OR

(iii) Find a relation between x and y such that the point (x, y) is equidistant from A and B.

38. A stadium has seats arranged in rows. The first row has 20 seats and each next row has 4 seats more than the row in front of it. The seat numbers in the rows form an AP.

(i) Find the first term a and common difference d of the AP. 1

(ii) Write the formula for the nth term of this AP. 1

(iii) How many seats are there in the 10th row? 2

OR

(iii) Find the total number of seats in the first 15 rows.